

TOPOLOGY

February 7, 2026

CONTENTS

The purpose of this document is to compile common notions in basic topology.

1. BASIC TOPOLOGY

1.1. Topological spaces.

Definition 1.1.1 (Topology). Let X be a set. A **topology on X** is a collection $\mathcal{T}$ of subsets of X such that:

1. $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$,
2. For any collection $\{U_i\}_{i \in I} \subseteq \mathcal{T}$ (with I arbitrary), the union $\bigcup_{i \in I} U_i \in \mathcal{T}$,
3. For any finite collection $\{U_1, \dots, U_n\} \subseteq \mathcal{T}$, the intersection $U_1 \cap \dots \cap U_n \in \mathcal{T}$.

If $\mathcal{T}$ is a topology on X , the pair $(X, \mathcal{T})$ is called a **topological space**. Members of $\mathcal{T}$ are called **open sets**.

A subset $C \subseteq X$ is **closed** if its complement $X \setminus C$ is an open set in $\mathcal{T}$.

One very often refers to X as a topological space, omitting the notation of the topology $\mathcal{T}$.

The collection of all topologies on a set X may be denoted by notations such as **Top(X)**, **Top(X)**, or **Top(X)**.

1.1.1. Basis of a topological space.

Definition 1.1.2. Let $(X, \mathcal{T})$ be a topological space and let $x \in X$.

- An **open neighborhood of x** is any open set $U \in \mathcal{T}$ such that $x \in U$.
- A **neighborhood of x** is a set $N \subseteq X$ for which there exists an open neighborhood $U \in \mathcal{T}$ of x such that $U \subseteq N$.
- A **neighborhood basis** (or **local base**) at x is a nonempty collection $\mathcal{B}_x$ of neighborhoods of x such that for every neighborhood N of x , there exists $B \in \mathcal{B}_x$ with $B \subseteq N$.

The elements of $\mathcal{B}_x$ are said to **form a base of neighborhoods** at x .

Definition 1.1.3. Let X be a set and let $\mathcal{B}$ be a collection of subsets of X . The collection $\mathcal{B}$ is called a **basis** (or **base**) for a topology on X if the following two conditions hold:

1. For every $x \in X$, there exists at least one $B \in \mathcal{B}$ such that $x \in B$.
2. For any $B_1, B_2 \in \mathcal{B}$ and any $x \in B_1 \cap B_2$, there exists $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

Given such a collection $\mathcal{B}$, the collection $\mathcal{T}$ of all unions of elements of $\mathcal{B}$ defines a topology on X , and it coincides with $\mathcal{T}_{\mathcal{B}}$, the topology generated by $\mathcal{B}$. In other words,

$$\mathcal{T}_{\mathcal{B}} = \{U \subseteq X : \text{for every } x \in U, \text{ there exists } B \in \mathcal{B} \text{ with } x \in B \subseteq U\}.$$

Definition 1.1.4. Let $(X, \mathcal{T})$ be a topological space.

- The space $(X, \mathcal{T})$ is said to be *first countable* if every point $x \in X$ has a countable neighborhood basis, i.e., there exists a countable collection $\{U_n\}_{n \in \mathbb{N}}$ of open neighborhoods of x such that for any open neighborhood U of x , there exists $n \in \mathbb{N}$ with $U_n \subseteq U$.
- The space $(X, \mathcal{T})$ is said to be *second countable* if there exists a countable basis $\mathcal{B} \subseteq \mathcal{T}$ for the topology $\mathcal{T}$, i.e., every open set $U \in \mathcal{T}$ can be written as a union of elements of $\mathcal{B}$.

1.1.2. Separation axioms.

Definition 1.1.5. Let $(X, \mathcal{T})$ be a topological space and let $A \subseteq X$.

- The set A is called a *G_{δ} set* if there exists a countable collection of open sets $\{U_n\}_{n \in \mathbb{N}} \subseteq \mathcal{T}$ such that

$$A = \bigcap_{n \in \mathbb{N}} U_n.$$

- The set A is called an *F_{σ} set* if there exists a countable collection of closed sets $\{F_n\}_{n \in \mathbb{N}}$ such that

$$A = \bigcup_{n \in \mathbb{N}} F_n.$$

Definition 1.1.6 (Separation axioms). Let $(X, \mathcal{T})$ be a topological space.

- $(X, \mathcal{T})$ is *T_0 (Kolmogorov)* if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that, without loss of generality, $x \in U$ and $y \notin U$.
- $(X, \mathcal{T})$ is *T_1 (Fréchet)* if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \in \mathcal{T}$ such that $x \in U$, $y \notin U$, and $y \in V$, $x \notin V$.
- $(X, \mathcal{T})$ is *T_2 or Hausdorff* if for every pair of distinct points $x, y \in X$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $x \in U$ and $y \in V$.
- $(X, \mathcal{T})$ is *regular* if it is T_1 and for each point $x \in X$ and closed set $F \subseteq X$ with $x \notin F$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $x \in U$ and $F \subseteq V$.
- $(X, \mathcal{T})$ is *T_3 (regular Hausdorff)* if it is T_1 and regular.
- $(X, \mathcal{T})$ is *completely regular* if for each closed set $F \subseteq X$ and $x \notin F$, there exists a continuous function $f : X \rightarrow [0, 1]$ such that $f(x) = 0$ and $f|_F = 1$.
- $(X, \mathcal{T})$ is *$T_{3\frac{1}{2}}$ (completely regular Hausdorff)* if it is T_1 and completely regular.
- $(X, \mathcal{T})$ is *normal* if it is T_1 and for each pair of disjoint closed sets $A, B \subseteq X$, there exist disjoint open sets $U, V \in \mathcal{T}$ such that $A \subseteq U$ and $B \subseteq V$.

- $(X, \mathcal{T})$ is T_4 (normal Hausdorff) if it is T_1 and normal.
- $(X, \mathcal{T})$ is T_5 (completely normal Hausdorff) if it is T_1 and completely normal.
- $(X, \mathcal{T})$ is *perfectly normal* if every closed set is a $G\delta$ (countable intersection of open sets) and the space is normal.
- $(X, \mathcal{T})$ is T_6 (perfectly normal Hausdorff) if it is T_1 and perfectly normal.

1.1.3. Subspaces of a topological space.

Definition 1.1.7 (Subspace topology and subspace). Let $(X, \mathcal{T})$ be a topological space and $Y \subseteq X$ a subset. The topology induced on Y by $\mathcal{T}$, called the *subspace topology*, is defined as

$$\mathcal{T}_Y := \{U \cap Y : U \in \mathcal{T}\}.$$

The pair $(Y, \mathcal{T}_Y)$ is called a *subspace of $(X, \mathcal{T})$* .

Convention 1.1.8. Given a topological space $(X, \mathcal{T})$ and a subset S of X , one almost always considers S as a subspace of X , i.e. considers S as a topological space equipped with the subspace topology.

Definition 1.1.9 (Discrete topological space). Let X be a set. The *discrete topology on X* is the topology $\mathcal{T}_{\text{disc}}$ defined by

$$\mathcal{T}_{\text{disc}} := \{U : U \subseteq X\},$$

i.e., every subset of X is open. The pair $(X, \mathcal{T}_{\text{disc}})$ is called a *discrete topological space*.

Definition 1.1.10 (Indiscrete (or trivial) topological space). Let X be a set. The *indiscrete topology* (also called the *trivial topology*) on X is the topology $\mathcal{T}_{\text{indisc}}$ defined by

$$\mathcal{T}_{\text{indisc}} := \{\emptyset, X\},$$

i.e., only the empty set and the entire set are open. The pair $(X, \mathcal{T}_{\text{indisc}})$ is called an *indiscrete topological space*.

Definition 1.1.11 (Closure of a subset). Let $(X, \mathcal{T})$ be a topological space and $A \subseteq X$ a subset. The *closure of A in X* , denoted by $\overline{A}$, is defined as the intersection of all closed sets containing A , i.e.,

$$\overline{A} := \bigcap \{C \subseteq X : C \text{ is closed and } A \subseteq C\}.$$

Equivalently, $\overline{A}$ is the smallest closed set containing A .

1.2. Properties of topological spaces.

Definition 1.2.1 (Dense subset). Let $(X, \mathcal{T})$ be a topological space. A subset $D \subseteq X$ is called *dense in X* if the closure of D equals X , i.e.,

$$\overline{D} = X.$$

Equivalently, D is dense in X if every non-empty open set $U \in \mathcal{T}$ intersects D , that is,

$$\forall U \in \mathcal{T}, U \neq \emptyset \implies U \cap D \neq \emptyset.$$

Definition 1.2.2 (Connected topological space). Let X be a topological space. The space X is called a *connected topological space*, or simply *connected*, if it cannot be written as a disjoint union $X = U \sqcup V$ of two nonempty open subsets $U, V \subseteq X$. Equivalently, X is connected if the only subsets of X that are both open and closed are $\emptyset$ and X itself.

1.2.1. Compact topological spaces.

Definition 1.2.3 (Compact topological space). A topological space $(X, \mathcal{T})$ is **compact** if every open cover of X admits a finite subcover; that is, for every collection $\{U_i\}_{i \in I}$ of open sets in $\mathcal{T}$ such that $X = \bigcup_{i \in I} U_i$, there exists a finite subcollection $\{U_{i_j}\}_{j=1}^n$ such that $X = \bigcup_{j=1}^n U_{i_j}$.

Some mathematicians, e.g. algebraic geometers, would refer to this property as **quasi-compactness**.

Here is an alternate, essentially equivalent definition, of compactness:

Definition 1.2.4 (Compactness). Let $(X, \mathcal{T})$ be a topological space. A subset $K \subseteq X$ is **compact** if for every collection $\{U_i\}_{i \in I}$ of open sets such that $K \subseteq \bigcup_{i \in I} U_i$, there exists a finite subcollection $\{U_{i_j}\}_{j=1}^n$ with $K \subseteq \bigcup_{j=1}^n U_{i_j}$.

Proposition 1.2.5. Let $(X, \mathcal{T})$ be a topological space. A subspace $K \subseteq X$ is compact in the sense of ?? if and only if the space K is compact in the sense of ??.

Definition 1.2.6 (Locally compact). Let $(X, \mathcal{T})$ be a topological space. X is **locally compact** if for every $x \in X$, there exists an open set $U \in \mathcal{T}$ containing x and a compact set $K \subseteq X$ such that $x \in U \subseteq K$.

Definition 1.2.7. Let $(X, \mathcal{T})$ be a topological space and let $A \subseteq X$.

- The **interior of A** , denoted by $\text{int}(A)$, is the union of all open sets contained in A :

$$\text{int}(A) = \bigcup \{U \in \mathcal{T} : U \subseteq A\}.$$

- The **boundary of A** , denoted by ∂A , is defined as

$$\partial A = \overline{A} \setminus \text{int}(A).$$

Equivalently, a point $x \in X$ belongs to ∂A if every open neighborhood of x intersects both A and its complement $X \setminus A$.

1.3. Continuous maps of topological spaces.

Definition 1.3.1. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. A map $f : X \rightarrow Y$ is called **continuous** if for every open set $V \in \mathcal{T}_Y$, the preimage $f^{-1}(V)$ is an open set in X , that is,

$$\forall V \in \mathcal{T}_Y, f^{-1}(V) \in \mathcal{T}_X.$$

Equivalently, f is continuous if and only if for every closed set $C \subseteq Y$, the preimage $f^{-1}(C)$ is closed in X .

A **map of topological spaces** usually refers to a continuous map between the topological spaces.

The set of continuous maps from X to Y is sometimes denoted by $C(X, Y)$. Other standard notation include $\text{Hom}_{\text{Top}}(X, Y)$ or $\text{Top}(X, Y)$ coming from more general notation for morphisms between objects in a category.

Definition 1.3.2. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. A function $f : X \rightarrow Y$ is called a **homeomorphism** if it satisfies all of the following:

1. f is bijective;
2. f is continuous with respect to $\mathcal{T}_X$ and $\mathcal{T}_Y$; and
3. the inverse map $f^{-1} : Y \rightarrow X$ is also continuous.

If such a function exists, the spaces X and Y are said to be **homeomorphic**.

Definition 1.3.3 (Pointed topological space). Let X be a topological space and let $x_0 \in X$ be a chosen element of X . A **pointed/based (topological) space** is a pair (X, x_0) consisting of the space X together with the distinguished point x_0 , called the **base point of X** . If the base point of a pointed space (X, x_0) is understood, then it may be suppressed from notation; in particular, X may be written as a pointed space as opposed to the full notation of (X, x_0) .

A **morphism of pointed spaces** (or **based map**) or **continuous map** between pointed spaces (X, x_0) and (Y, y_0) is a continuous map

$$f : X \rightarrow Y$$

such that $f(x_0) = y_0$.

The collection of pointed spaces with their morphisms form a locally small category, often called the **category of pointed spaces**. This category is often denoted by notations such as **Top_{*}**, **Top_•**, **Top_{*}**, **Top_•**, etc. The set of continuous maps from pointed spaces X to Y may denoted by notations such as **C_{*}(X, Y)**, **C_•(X, Y)**, **Top_{*}(X, Y)**, **Top_•(X, Y)**, **Hom_{Top_•}(X, Y)**, etc.

Definition 1.3.4 (Preimage of a subset under a map of sets). Let X and Y be sets and let $f : X \rightarrow Y$ be a function. Let $B \subseteq Y$ be a subset of the codomain Y . The **preimage of B under f** (also called the **inverse image of B under f**) is the subset of X defined by

$$f^{-1}(B) := \{x \in X : f(x) \in B\} \subseteq X.$$

If B is a singleton set $\{y\}$, then we often denote $f^{-1}(B)$ by **$f^{-1}(y)$** .

Definition 1.3.5 (Fiber of a map of topological spaces). Let X and Y be topological spaces and let $f : X \rightarrow Y$ be a continuous map. For a point $y \in Y$, the **fiber of f over y** is the inverse image $f^{-1}(y) = f^{-1}(\{y\})$ endowed with the subspace topology induced from X . The fiber is also denoted by notations such as **Fib _{f} (y)** or **X_y** .

1.4. Constructions of topological spaces from other topological spaces.

Definition 1.4.1 (Product of Sets). Let I be a (possibly infinite but small) index set and let $\{A_i\}_{i \in I}$ be a family of sets indexed by I . The **Cartesian product of the family $\{A_i\}_{i \in I}$** , denoted by **$\prod_{i \in I} A_i$** , is defined as the set of all tuples/functions

$$\prod_{i \in I} A_i := \{(a_i)_{i \in I} \mid a_i \in A_i \text{ for all } i \in I\},$$

where $(a_i)_{i \in I}$ denotes a function from I to $\bigcup_{i \in I} A_i$ such that $(a_i)_{i \in I}(i) = a_i \in A_i$ for each $i \in I$.

The self product of a set A indexed by I is often denoted by A^I . Note that elements of A^I can be identified with functions $I \rightarrow A$. The finite self product of A taken n times is often denoted by A^n . For finitely many sets $A_1, \dots, A_n$, their Cartesian product is denoted by $A_1 \times \dots \times A_n$. Elements of such a finite product may be written as $(a_1, \dots, a_n)$.

Definition 1.4.2. Let I be an index set and $\{X_i\}_{i \in I}$ be a collection of topological spaces. Let $X = \prod_{i \in I} X_i$ be the set theoretic Cartesian product of these spaces.

The *box topology on X* is the topology generated by the base consisting of all sets of the form $\prod_{i \in I} U_i$, where each U_i is an open subset of X_i .

Definition 1.4.3. Let I be an index set and $\{X_i\}_{i \in I}$ be a collection of topological spaces. Let $X = \prod_{i \in I} X_i$ be the set theoretic Cartesian product of these spaces.

The *product topology* (or *Tychonoff topology*) on X is the topology generated by the base consisting of all sets of the form $\prod_{i \in I} U_i$, where each U_i is an open subset of X_i and $U_i = X_i$ for all but finitely many indices $i \in I$.

The *product* or the *product space of the X_i* refers to X equipped with the product topology.

For each $j \in I$, let $p_j : \prod_{i \in I} X_i \rightarrow X_j$ denote the canonical projection map. The product topology is the coarsest topology on X such that all projection maps p_j are continuous.

Proposition 1.4.4. Let I be an index set and $\{X_i\}_{i \in I}$ be a collection of topological spaces. Let $X = \prod_{i \in I} X_i$ be the set theoretic Cartesian product of these spaces.

If $X = \prod_{i \in I} X_i$ is endowed with the product topology, then a function $f : Z \rightarrow X$ from a topological space Z is continuous if and only if each component function $f_i = p_i \circ f : Z \rightarrow X_i$ is continuous.

This universal property generally fails for the box topology when I is infinite.

Theorem 1.4.5 (Tychonoff's theorem). Let I be an index set and let $\{X_i\}_{i \in I}$ be a collection of compact spaces.

The product space $X = \prod_{i \in I} X_i$ is compact in the product topology.

1.5. Compactly generated.

Definition 1.5.1. (♠ TODO: final topology) A topological space X is said to be *compactly generated* (or a *k-space*) if a subset $U \subseteq X$ is open whenever for every compact subset $K \subseteq X$, the intersection $U \cap K$ is open in the subspace K . Equivalently, X is compactly generated if and only if the topology of X is the final topology with respect to the collection of inclusions $K \hookrightarrow X$ for compact $K \subseteq X$.

Definition 1.5.2. A topological space X is said to be *weakly Hausdorff* if for every continuous map $f : K \rightarrow X$ from a compact Hausdorff space K , the image $\text{Im}(f)$ is closed in X . Equivalently, X is weakly Hausdorff if and only if for every compact Hausdorff K , the induced map $f : K \rightarrow X$ is a closed map.

1.6. Metrics, absolute values, and norms.

1.6.1. Extended metric spaces.

Definition 1.6.1 (Extended Metric). Let M be a set. An *extended metric* on M is a function

$$d : M \times M \rightarrow [0, \infty]$$

such that for all $x, y, z \in M$:

1. **Non-negativity:** $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$.
2. **Symmetry:** $d(x, y) = d(y, x)$.
3. **Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$,

again adopting the convention that sums involving ∞ behave so that $a + \infty = \infty$. A set equipped with an extended metric is called an *extended metric space*.

If the range of the extended metric is contained in $[0, \infty)$, then the extended metric is a *metric* in the usual sense and V may be called a *metric space*.

Definition 1.6.2 (Isometry for extended metric spaces). Let (X, d_X) and (Y, d_Y) be extended metric spaces. A map $f : X \rightarrow Y$ is called an *isometry* if for all $x_1, x_2 \in X$,

$$d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2).$$

An isometry is automatically injective. If, in addition, f is surjective, it is called an *isometric isomorphism*, and X and Y are said to be *isometric*.

Definition 1.6.3 (Extended Metric Induced by an Extended Norm). Let V be a vector space over a field F equipped with an absolute value

$$|\cdot| : F \rightarrow [0, \infty),$$

and let $\|\cdot\| : V \rightarrow [0, \infty]$ be an extended norm on V . Then the *extended metric induced by the extended norm* is the function

$$d : V \times V \rightarrow [0, \infty]$$

defined by

$$d(x, y) := \|x - y\|.$$

It is indeed an extended metric. If $\|\cdot\|$ is a norm, then d is a metric.

Definition 1.6.4 (Topology induced by an extended metric). Let (X, d) be an extended metric space.

1. Given $\varepsilon \in (0, \infty)$ and $x \in X$, the set

$$B(x, \varepsilon) := \{y \in X : d(x, y) < \varepsilon\}$$

is called the *open ball of radius ε centered at x* .

2. The *topology induced by d on X* is the topology defined by declaring a subset $U \subseteq X$ to be open if for every $x \in U$, there exists $\varepsilon \in (0, \infty)$ such that $B(x, \varepsilon)$ is contained in U .

1.7. Convergence in extended metric spaces.

Definition 1.7.1 (Convergence and Limits in an Extended Metric Space). Let (X, d) be an extended metric space. A sequence $(x_n)_{n=1}^{\infty}$ in X *converges to a point* $x \in X$ if for every $\varepsilon > 0$, there exists an integer $N \in \mathbb{N}$ such that for all $n \geq N$,

$$d(x_n, x) < \varepsilon.$$

In that case, x is called the *limit of the sequence* (x_n) , and we write

$$\lim_{n \rightarrow \infty} x_n = x \quad \text{or} \quad x_n \rightarrow x$$

Definition 1.7.2 (Cauchy sequence in an extended metric space). Let (M, d) be an extended metric space. A sequence $(x_n)_{n \in \mathbb{N}}$ in M is called a *Cauchy sequence* if for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$, $d(x_m, x_n) < \epsilon$ (and thus, in particular, all pairwise distances $d(x_m, x_n)$ are finite for $m, n \geq N$).

Definition 1.7.3. Let (X, d) be an extended metric space. The space (X, d) is said to be *complete* if every Cauchy sequence in X converges to a point in X ; that is, for every sequence $(x_n)_{n \in \mathbb{N}}$ in X such that for all $\varepsilon > 0$ there exists $N \in \mathbb{N}$ with $d(x_m, x_n) < \varepsilon$ for all $m, n \geq N$, there exists $x \in X$ such that

$$\lim_{n \rightarrow \infty} d(x_n, x) = 0.$$

Definition 1.7.4. Let (X, d) be an extended metric space. A *completion of* (X, d) is a pair $(\widehat{X}, \widehat{d})$ together with an isometric embedding $i : X \rightarrow \widehat{X}$, where:

- $(\widehat{X}, \widehat{d})$ is a complete extended metric space,
- $i(X)$ is dense in $\widehat{X}$ (with respect to the topologies induced by the extended metrics),
- $d(x, y) = \widehat{d}(i(x), i(y))$ for all $x, y \in X$.

(♠ TODO: prove that completions of extended metric spaces exist) As such a space $\widehat{X}$ exist, and is unique up to unique isometry, we may refer to the space $\widehat{X}$ as called *the completion of* X (with respect to d).

Proposition 1.7.5. Let (X, d) be an extended metric space. A completion $(\widehat{X}, \widehat{d})$ of (X, d) exists and is unique up to unique isometry. Moreover, if (X, d) is a metric space, then so is $(\widehat{X}, \widehat{d})$.

Proof. (♠ TODO: :) actually complete this proof Let $\mathcal{C}$ be the set of all Cauchy sequences in (X, d) :

$$\mathcal{C} = \{(x_n)_{n \in \mathbb{N}} : (x_n) \text{ is a Cauchy sequence in } (X, d)\},$$

and define a relation $\sim$ on $\mathcal{C}$ by

$$(x_n) \sim (y_n) \iff \lim_{n \rightarrow \infty} d(x_n, y_n) = 0.$$

This relation is well-defined and in fact an equivalence relation. (♠ TODO: Show that the relation is well-defined and an equivalence relation) Let $\widehat{X} = \mathcal{C} / \sim$ and let $\widehat{d} : \widehat{X} \times \widehat{X} \rightarrow [0, \infty]$ be defined by

$$\widehat{d}([(x_n)], [(y_n)]) := \lim_{n \rightarrow \infty} d(x_n, y_n).$$

This limit exists as a value in $[0, \infty]$ and is independent of representative sequences chosen.
 (♠ TODO: Demonstrate why the limit exists and is well defined)

It can be verified that $\widehat{d}$ is an extended metric on $\widehat{X}$ and that $(\widehat{X}, \widehat{d})$ is complete by construction.

Define an embedding

$$i : X \rightarrow \widehat{X}, \quad i(x) := [(x, x, x, \dots)],$$

which is an isometry:

$$\widehat{d}(i(x), i(y)) = \lim_{n \rightarrow \infty} d(x, y) = d(x, y).$$

Moreover, $i(X)$ is dense in $\widehat{X}$ because any equivalence class $[(x_n)]$ of a Cauchy sequence can be approximated arbitrarily closely by points $i(x_n)$ in the image of X .

Uniqueness:

Suppose $(\widehat{X}', \widehat{d}')$ is another completion of (X, d) with isometric embedding $i' : X \rightarrow \widehat{X}'$.

By the universal property of completions, there exists a unique isometry

$$\Phi : \widehat{X} \rightarrow \widehat{X}'$$

such that $\Phi \circ i = i'$.

The map Φ is onto since $i'(X)$ is dense in $\widehat{X}'$ and Φ preserves limits of Cauchy sequences.

Therefore, Φ is a unique isometry between the completions, and the completion is unique up to unique isometry. $\square$

1.7.1. Absolute values on fields.

Definition 1.7.6 (Absolute Value on a Field). Let F be a field. An *absolute value on F* is a function

$$|\cdot| : F \rightarrow \mathbb{R}_{\geq 0}$$

that satisfies the following properties for all $a, b \in F$:

1. Non-negativity: $|a| \geq 0$,
2. Positive-definiteness: $|a| = 0 \iff a = 0$,
3. Multiplicativity: $|ab| = |a| \cdot |b|$,
4. Triangle inequality: $|a + b| \leq |a| + |b|$.

Here, 0 denotes the additive identity of the field F , and the codomain $\mathbb{R}_{\geq 0}$ consists of non-negative real numbers.

1.7.2. Norms on vector spaces over fields with absolute values.

Definition 1.7.7 (Extended Norm). Let V be a vector space over a field F equipped with an absolute value

$$|\cdot| : F \rightarrow [0, \infty).$$

An **extended norm on V** is a function

$$\|\cdot\| : V \rightarrow [0, \infty]$$

satisfying for all $x, y \in V$ and all scalars $\alpha \in F$:

1. **Positive definiteness:** $\|x\| = 0$ if and only if $x = 0$.
2. **Homogeneity:** $\|\alpha x\| = |\alpha| \cdot \|x\|$.
3. **Triangle inequality:** $\|x + y\| \leq \|x\| + \|y\|$,

where arithmetic is extended to allow sums involving ∞ with the convention that $a + \infty = \infty$ for any $a \in [0, \infty]$. A vector space with an extended norm over a field with an absolute value is called an **extended normed (vector) space**.

If the range of the extended norm is contained in $[0, \infty)$, then the extended norm is a **norm** in the usual sense and V may be called a **normed (vector) space**.

Definition 1.7.8 (Topology induced by a norm). Let V be a vector space over a field K with absolute value $|\cdot|$, and let $\|\cdot\|$ be an extended norm on V . The **topology induced by the extended norm $\|\cdot\|$ on V** is defined by declaring a subset $U \subseteq V$ to be open if for every $x \in U$, there exists $\varepsilon > 0$ such that

$$B(x, \varepsilon) := \{y \in V : \|y - x\| < \varepsilon\}$$

is contained in U . The set $B(x, \varepsilon)$ is called the **open ball of radius ε around x** . The collection of all such open sets forms a topology on V .

1.7.3. Inner product on a vector space over the reals or complex numbers.

Definition 1.7.9. An **inner product on a vector space V over a field $\mathbb{F}$** (either $\mathbb{R}$ or $\mathbb{C}$) is a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$$

satisfying, for all $u, v, w \in V$ and all scalars $\alpha \in \mathbb{F}$:

1. **Conjugate symmetry:** $\langle u, v \rangle = \overline{\langle v, u \rangle}$
2. **Linearity in the first argument:** $\langle \alpha u + v, w \rangle = \alpha \langle u, w \rangle + \langle v, w \rangle$
3. **Positive-definiteness:** $\langle v, v \rangle \geq 0$ and $\langle v, v \rangle = 0$ if and only if $v = 0$

A vector space over $\mathbb{F}$ equipped with such an inner product is called an **inner product space**.

Definition 1.7.10. Let V be a vector space over a field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let $\langle \cdot, \cdot \rangle$ be an inner product. The **norm induced by $\langle \cdot, \cdot \rangle$** is the norm

$$\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$$

defined by

$$\|v\| = \sqrt{\langle v, v \rangle}$$

for any $v \in V$.

1.8. Euclidean space.

Definition 1.8.1. For a positive integer n , let $\mathbb{R}^n$ denote the n -fold Cartesian product of the real line $\mathbb{R}$ with itself:

$$\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{R} \text{ for all } i = 1, \dots, n\}.$$

The set $\mathbb{R}^n$ is called **Euclidean n-space**. A point $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ is associated with the **Euclidean norm**

$$\|x\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}.$$

The corresponding metric $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow [0, \infty)$ is given by

$$d(x, y) = \|x - y\|.$$

This metric induces the standard topology on $\mathbb{R}^n$, called the *Euclidean topology*.

2. HOMOTOPY

The notions of homotopy have the following variants:

1. Absolute, on unpointed spaces
2. Relative, on unpointed spaces
3. Absolute, on pointed spaces
4. Relative, on pointed spaces.

Definition 2.0.1 (Homotopy of maps of topological spaces). Let X and Y be topological spaces and let $K \subseteq X$ be a subset. Let $C(X, Y)$ denote the set of all continuous maps $f : X \rightarrow Y$.

1. A *homotopy between two maps $f, g \in C(X, Y)$ relative to K* is a continuous map

$$H : X \times [0, 1] \rightarrow Y$$

such that for all $x \in X$,

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and for all $x \in K$ and $t \in [0, 1]$,

$$H(x, t) = f(x) = g(x).$$

If such an H exists, we say f and g are *homotopic relative to K* , and we write $f \simeq g \text{ rel } K$; this is an equivalence relation.

A *homotopy between two maps $f, g \in C(X, Y)$* is simply a homotopy relative to $\emptyset$. We write we write $f \simeq g$ if a homotopy between them exists.

- Let (X, x_0) and (Y, y_0) be pointed topological spaces and let $K \subseteq X$ be a subset with $x_0 \in K$. Let $C_*(X, Y)$ denote the set of all continuous based maps $f : X \rightarrow Y$ satisfying $f(x_0) = y_0$.

A *homotopy of based maps $f, g \in C_*(X, Y)$ relative to K* is a continuous map

$$H : X \times [0, 1] \rightarrow Y$$

such that for all $x \in X$,

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and for all $k \in K$ and $t \in [0, 1]$,

$$H(k, t) = f(k) = g(k),$$

in particular fixing the basepoint throughout,

$$H(x_0, t) = y_0 \quad \text{for all } t \in [0, 1].$$

If such an H exists, we say f and g are *based homotopic relative to K* , and we write $f \simeq g \text{ rel } K$. This is an equivalence relation.

A *homotopy of based maps $f, g \in C_*(X, Y)$* without relative condition is the special case $K = \{x_0\}$ and is called a *homotopy of based maps* or *based homotopy*. We write $f \simeq g$ if such a homotopy exists.

Definition 2.0.2 (Homotopy equivalence). Let X and Y be topological spaces and let $K \subseteq X$.

- A continuous map $f : X \rightarrow Y$ is a *homotopy equivalence relative to K* if there exists a continuous map $g : Y \rightarrow X$ such that

$$g \circ f \simeq \text{id}_X \text{ rel } K \quad \text{and} \quad f \circ g \simeq \text{id}_Y \text{ rel } f(K).$$

(??) That is, there exist homotopies $H : X \times [0, 1] \rightarrow X$ and $G : Y \times [0, 1] \rightarrow Y$ satisfying

$$\begin{aligned} H(x, 0) &= (g \circ f)(x), & H(x, 1) &= x, & H(k, t) &= k \text{ for all } k \in K, t \in [0, 1], \\ G(y, 0) &= (f \circ g)(y), & G(y, 1) &= y, & G(f(k), t) &= f(k) \text{ for all } k \in K, t \in [0, 1]. \end{aligned}$$

In this case, the map g is called a *homotopy inverse relative to K of f* , and the pairs (X, K) and $(Y, f(K))$ are said to be *homotopy equivalent relative to K* , denoted $(X, K) \simeq (Y, f(K)) \text{ rel } K$. This is an equivalence relation.

A continuous map $f : X \rightarrow Y$ is a *homotopy equivalence* if it is a homotopy equivalence relative to $\emptyset$. A *homotopy inverse of f* is then simply a homotopy inverse of f relative to $\emptyset$. In this case, and the spaces X and Y are said to be *homotopy equivalent*, denoted $X \simeq Y$. This is an equivalence relation.

- Let (X, x_0) and (Y, y_0) be pointed topological spaces and let $K \subseteq X$ be a subset with $x_0 \in K$.

A continuous map $f : (X, x_0) \rightarrow (Y, y_0)$ of pointed spaces is a *homotopy equivalence relative to K* if there exists a continuous map $g : (Y, y_0) \rightarrow (X, x_0)$ such that

$$g \circ f \simeq \text{id}_X \text{ rel } K \quad \text{and} \quad f \circ g \simeq \text{id}_Y \text{ rel } f(K),$$

where the homotopies are homotopies of based maps relative to K and $f(K)$, respectively.

That is, there exist homotopies of based maps $H : X \times [0, 1] \rightarrow X$ and $G : Y \times [0, 1] \rightarrow Y$ satisfying

$$\begin{aligned} H(x, 0) &= (g \circ f)(x), & H(x, 1) &= x, & H(k, t) &= k \text{ for all } k \in K, t \in [0, 1], \\ G(y, 0) &= (f \circ g)(y), & G(y, 1) &= y, & G(f(k), t) &= f(k) \text{ for all } k \in K, t \in [0, 1], \end{aligned}$$

with $H(x_0, t) = x_0$ and $G(y_0, t) = y_0$ for all t .

The map g is called a *homotopy inverse relative to K of f* , and the pairs (X, K) and $(Y, f(K))$ are said to be *homotopy equivalent relative to K* , denoted $(X, K) \simeq (Y, f(K)) \text{ rel } K$. This is an equivalence relation.

A continuous map $f : (X, x_0) \rightarrow (Y, y_0)$ of pointed spaces is a *homotopy equivalence* if it is a homotopy equivalence relative to $\{x_0\}$. A *homotopy inverse of f* is then simply a homotopy inverse relative to $\{x_0\}$. In this case, the pointed spaces (X, x_0) and (Y, y_0) are said to be *homotopy equivalent*, denoted $(X, x_0) \simeq (Y, y_0)$.

Definition 2.0.3 (Homotopy class of maps relative to a subset). Let X and Y be topological spaces and let $K \subseteq X$. Let $C(X, Y)$ denote the set of all continuous maps $f : X \rightarrow Y$.

- Two maps $f, g \in C(X, Y)$ are said to be in the same *homotopy class relative to K* if there exists a homotopy relative to K

$$H : X \times [0, 1] \rightarrow Y$$

such that

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and

$$H(k, t) = f(k) = g(k) \quad \text{for all } k \in K, t \in [0, 1].$$

The *homotopy class of maps relative to K* containing a map $f : X \rightarrow Y$ is denoted by $[f]_K$.

Two maps $f, g \in C(X, Y)$ are said to be in the same *homotopy class* if they are in the same homotopy class relative to $\emptyset$.

The *homotopy class of maps* containing a map $f : X \rightarrow Y$ is denoted by $[f]$.

The set of homotopy classes of maps may often be denoted by $[X, Y]$.

- Let (X, x_0) and (Y, y_0) be pointed topological spaces and let $K \subseteq X$ be a subset containing x_0 . Let $C_*(X, Y)$ denote the set of all continuous based maps $f : X \rightarrow Y$ with $f(x_0) = y_0$.

Two based maps $f, g \in C_*(X, Y)$ are said to be in the same *homotopy class relative to K* if there exists a homotopy of based maps relative to K

$$H : X \times [0, 1] \rightarrow Y$$

such that for all $x \in X$,

$$H(x, 0) = f(x), \quad H(x, 1) = g(x),$$

and for all $k \in K$ and $t \in [0, 1]$,

$$H(k, t) = f(k) = g(k),$$

particularly ensuring the basepoint is fixed throughout,

$$H(x_0, t) = y_0 \quad \text{for all } t \in [0, 1].$$

The *homotopy class relative to K* containing $f : (X, x_0) \rightarrow (Y, y_0)$ is denoted by $[f]_K$.

Two based maps $f, g \in C_*(X, Y)$ are said to be in the same *homotopy class* if they are in the same homotopy class relative to $\{x_0\}$.

The *homotopy class* containing a map $f : (X, x_0) \rightarrow (Y, y_0)$ is denoted by $[f]$.

The set of homotopy classes of pointed maps $(X, x_0) \rightarrow (Y, y_0)$ may often be denoted by $[(X, x_0), (Y, y_0)]$ or by $[X, Y]$ if the base points are clear.

Definition 2.0.4. The *homotopy category of topological spaces*, denoted $\mathbf{hTop}$, is the category whose objects are topological spaces and whose morphisms are homotopy classes of continuous maps. In other words, for objects X and Y , the set of morphisms is defined as $\text{Hom}_{\mathbf{hTop}}(X, Y) = [X, Y] = C(X, Y)/\simeq$.

Proposition 2.0.5. Composition in the homotopy category of topological spaces is well-defined. If $f_1, f_2 : X \rightarrow Y$ are homotopic and $g_1, g_2 : Y \rightarrow Z$ are homotopic, then the compositions $g_1 \circ f_1$ and $g_2 \circ f_2$ are homotopic as maps from X to Z . That is, $[g] \circ [f] = [g \circ f]$ is independent of the choice of representatives.

Theorem 2.0.6. There exists a canonical functor $Q : \mathbf{Top} \rightarrow \mathbf{hTop}$ which is the identity on objects and maps each continuous map f to its homotopy class $[f]$. This functor is full and essentially surjective.

Definition 2.0.7 (Path in a topological space). Let X be a topological space, let $x_0, x_1 \in X$, and let $[0, 1] \subset \mathbb{R}$ be the closed unit interval equipped with the subspace topology from $\mathbb{R}$.

1. A *path in X from x_0 to x_1* is a continuous map $\gamma : [0, 1] \rightarrow X$ such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$. More generally, a *path in X* is a path from some point $x_0 \in X$ to some point $x_1 \in X$ in this sense.
2. A *loop in X based at $x_0 \in X$* is a path $\gamma : [0, 1] \rightarrow X$ in X from x_0 to x_0 , that is, a continuous map γ such that $\gamma(0) = \gamma(1) = x_0$. When a basepoint x_0 is fixed and understood from the context, a *loop in X* means a loop in X based at this chosen basepoint.

Definition 2.0.8 (Homotopy groups). For any pointed topological space (X, x_0) and integer $n \geq 0$, the *n -th homotopy group of X at x_0* , denoted $\pi_n(X, x_0)$, is defined as the set of all homotopy classes (rel. ∂I^n) of based maps

$$f : (I^n, \partial I^n) \rightarrow (X, x_0),$$

where $I^n = [0, 1]^n$. For $n \geq 1$, $\pi_n(X, x_0)$ is a group under concatenation of based maps, and for $n \geq 2$, it is abelian.

The *fundamental group of (X, x_0)* refers to $\pi_1(X, x_0)$. Equivalently, it is the group of homotopy classes (rel. endpoints) of loops $\gamma : [0, 1] \rightarrow X$ satisfying $\gamma(0) = \gamma(1) = x_0$.

Definition 2.0.9 (Weak homotopy equivalence). Let $f : X \rightarrow Y$ be a continuous map between topological spaces. The map f is a *weak homotopy equivalence* if, for every choice of basepoint $x_0 \in X$ with $y_0 = f(x_0)$, the induced maps on homotopy groups

$$f_* : \pi_n(X, x_0) \rightarrow \pi_n(Y, y_0)$$

are isomorphisms for all integers $n \geq 0$.

3. SINGULAR HOMOLOGY AND COHOMOLOGY

Definition 3.0.1. Let V be a real vector space of finite dimension. A *k -simplex in topology* (or a *geometric k -simplex*) is the convex hull of $k + 1$ affinely independent points $v_0, v_1, \dots, v_k \in V$, and is denoted by

$$[v_0, v_1, \dots, v_k] := \left\{ \sum_{i=0}^k t_i v_i \mid t_i \geq 0, \sum_{i=0}^k t_i = 1 \right\}.$$

It is also standard to talk of the *standard topological n -simplex* — the topological space, often denoted by notations such as Δ^n or $|\Delta^n|$ defined as the subset of Euclidean space $\mathbb{R}^{n+1}$ given by

$$|\Delta^n| = \left\{ (t_0, t_1, \dots, t_n) \in \mathbb{R}^{n+1} : \sum_{i=0}^n t_i = 1, \text{ and } t_i \geq 0 \text{ for all } i \right\}$$

equipped with the induced topology from the usual Euclidean topology on $\mathbb{R}^{n+1}$.

(♠ TODO: comment on how $|\Delta^n|$ makes sense via a geometric realization)

Definition 3.0.2 (Singular simplices). Let X be a topological space. For each integer $n \geq 0$, A *singular n -simplex in X* is a continuous map

$$\sigma : \Delta^n \rightarrow X.$$

where Δ^n is the standard topological n -simplex. The set of all singular n -simplices in X is denoted by $S_n(X)$.

Definition 3.0.3 (Singular chain group with coefficients). Let X be a topological space, let $A \subseteq X$ be a subspace, and let $S_n(X)$ be the set of singular n -simplices in X .

1. Let R be a ring.
 - (a) The *singular n -chain group of X with coefficients in R* is the free R -module $C_n(X; R)$ whose elements are finite formal linear combinations

$$\sum_i r_i \sigma_i, \quad \text{with } \sigma_i \in S_n(X), r_i \in R.$$

Elements of $C_n(X; R)$ are called *singular n -chains in X with coefficients in R* .

- (b) If $A \subseteq X$ is a subspace, the quotient groups

$$C_n(X, A; R) = C_n(X; R) / C_n(A; R)$$

may be referred to as the *relative singular n -chain groups of X in A with coefficients in R* and elements of this group may be referred to as *relative singular n -chains in X relative to A with coefficients in R* .

In either case, when $R = \mathbb{Z}$, the ring R may be suppressed from notation, so we may write $C_n(X)$ and $C_n(X, A)$ for $C_n(X, \mathbb{Z})$ and $C_n(X, A, \mathbb{Z})$ respectively.

2. Let G be an abelian group.

- (a) More generally, the *singular n -chain group of X with coefficients in G* is the abelian group

$$C_n(X; G) = C_n(X; \mathbb{Z}) \otimes_{\mathbb{Z}} G.$$

(??) In particular, an element of $C_n(X; G)$ is a finite formal sum

$$\sum_i g_i \sigma_i, \quad g_i \in G, \sigma_i \in S_n(X).$$

Elements of $C_n(X; G)$ are called *singular n -chains in X with coefficients in G* .

- (b) Similarly, the *relative singular n -chain group of X in A with coefficients in G* is the quotient group

$$C_n(X, A; G) = C_n(X; G) / C_n(A; G)$$

Elements of $C_n(X, A; G)$ are called *singular n -chains in X relative to A with coefficients in G* .

When $G = R$ is a ring, note that the notions of $C_n(X; R)$ where R is taken as a ring and where R is taken as an abelian group are equivalent; similarly for $C_n(X, A; R)$.

We note that, in general $C_n(X; G) \cong C_n(X, \emptyset; G)$.

Definition 3.0.4 (Singular chain complex with coefficients). Let X be a topological space, let $A \subseteq X$ be a subspace, and let G be an abelian group.

For each $n \geq 1$, define the *boundary operator*

$$\partial_n : C_n(X; G) \rightarrow C_{n-1}(X; G)$$

by

$$\partial_n(\sigma) = \sum_{i=0}^n (-1)^i \sigma \circ \delta_i,$$

where $\delta_i : \Delta^{n-1} \rightarrow \Delta^n$ is the i -th face inclusion, and extend ∂_n to $C_n(X; G)$ linearly. For $n \leq 0$, we simply let $\partial_n = 0$

One can verify that $(C_n(X; G), \partial_n)$ forms a chain complex in the abelian category of abelian groups. This chain complex is called the *(absolute) singular chain complex of X with coefficients in G* and is often denoted by $C_{\bullet}(X; G)$.

Then the boundary maps above induce maps

$$C_n(X, A; G) \rightarrow C_{n-1}(X, A; G)$$

on the relative chain groups $C_n(X, A; G)$, yielding a chain complex of abelian groups whose boundary maps, often also denoted by ∂_n , are induced by the boundary operators ∂_n above; this chain complex may be called the *relative singular chain complex of the pair (X, A) with coefficients in G* and is often denoted by $C_{\bullet}(X, A; G)$.

In the case that $G = R$ is a (not necessarily commutative) ring, then the boundary maps of $C_{\bullet}(X; R)$ and $C_{\bullet}(X, A; R)$ are left R -module homomorphisms:

$$\partial \left(\sum_i r_i \sigma_i \right) = \sum_i r_i \partial(\sigma_i)$$

Definition 3.0.5 (Singular homology with coefficients). Let X be a topological space, let $A \subseteq X$ be a subspace, and let G be an abelian group.

The *(absolute) n -th singular homology group of X with coefficients in G* is the homology group

$$H_n(X; G) = H_n(C_*(X; G)) = \ker \partial_n / \operatorname{im} \partial_{n+1}$$

where $C_*(X; G)$ is the singular chain complex of X with coefficients in G and $\partial_n : C_n(X; G) \rightarrow C_{n-1}(X; G)$ are the boundary maps.

Given a subspace $A \subseteq X$, the *n -th relative singular homology group of (X, A) with coefficients in G* is defined as

$$H_n(X, A; G) = H_n(C_*(X, A; G)) = \ker \partial_n / \operatorname{im} \partial_{n+1}$$

where $C_*(X, A; G)$ is the relative singular chain complex of (X, A) with coefficients in G and $\partial_n : C_n(X, A; G) \rightarrow C_{n-1}(X, A; G)$ are the boundary maps.

We may denote $H_n(X; \mathbb{Z})$ and $H_n(X, A; \mathbb{Z})$ by $H_n(X)$ and $H_n(X, A)$ respectively.

In the case that $G = R$ is a ring, $H_n(X; R)$ and $H_n(X, A; R)$ have the structure of left R -modules since the corresponding chain complexes are chain complexes of left R -modules..

Definition 3.0.6 (Singular cochain complex of a topological space). Let X be a topological space and R a commutative ring with 1. The *singular n -cochain group of X with coefficients in R* is the R -module

$$C^n(X; R) = \operatorname{Hom}_R(C_n(X; R), R).$$

We define *coboundary operators*

$$\delta^n : C^n(X; R) \rightarrow C^{n+1}(X; R), \quad \delta^n(\varphi) = \varphi \circ \partial_{n+1}.$$

The pair $(C^n(X; R), \delta^n)_n$ form a cochain complex called the *singular cochain complex of X with coefficients in R* . Given a subspace $A \subseteq X$, the *relative singular n -cochain group of (X, A) with coefficients in R* is the R -module

$$C^n(X, A; R) = \operatorname{Hom}_R(C_n(X, A; R), R).$$

The coboundary operators δ^n induce maps $C^n(X, A; R) \rightarrow C^{n+1}(X, A; R)$ for which the $C^n(X, A; R)$ form a cochain complex, called the *relative singular cochain complex of (X, A) with coefficients in R*

Definition 3.0.7. Let X be a topological space and R a commutative ring with 1. The *n -th singular cohomology group of X with coefficients in R* is

$$H^n(X; R) = \ker(\delta^n) / \operatorname{im}(\delta^{n-1}).$$

For a subspace $A \subseteq X$ the *relative cohomology groups of (X, A)* are

$$H^n(X, A; R) = H^n(C^*(X, A; R)).$$

In other words, these singular cohomology groups are the cohomology R -modules of the singular cochain complex $C^*(X; R)$ and the relative singular cochain complex $C^*(X, A; R)$ respectively.

APPENDIX A. SET THEORY

Definition A.0.1. Let A be a set.

- The set A is said to be *countably infinite* (or simply *countable*) if there exists a bijection $f : \mathbb{N} \rightarrow A$.
- The set A is said to be *finite* if there exists some $n \in \mathbb{N}$ and a bijection $g : \{1, 2, \dots, n\} \rightarrow A$.
- The set A is said to be *at most countable* if it is either finite or countably infinite.
- The set A is said to be *uncountable* if it is not at most countable.

Definition A.0.2. Let X and Y be sets and let $f : X \rightarrow Y$ be a function.

- The function f is said to be *injective* (or *one-to-one*) if for all $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
- The function f is said to be *surjective* (or *onto*) if for every $y \in Y$ there exists $x \in X$ such that $f(x) = y$.
- The map f is *bijective* if it is both injective and surjective. In this case, there exists a unique *inverse map* $f^{-1} : Y \rightarrow X$ such that for all $x \in X$ and $y \in Y$,

$$f^{-1}(f(x)) = x \text{ and } f(f^{-1}(y)) = y.$$

APPENDIX B. CATEGORY THEORY

Definition B.0.1 (Category). A category name=Category, description=A nice enough collection of objects and morphisms (??), *category* category $\mathcal{C}$ consists of the following data:

- A class of object_{of}categoryname = *Objectofacategory*, description = ??, *format* = *textbf* *objectofacategorydenotedclassofobjects* *of*categoryname = $\text{Ob}(\mathcal{C})$, description=Class of objects of a category $\mathcal{C}$??, sort=Ob, $\text{Ob}(\mathcal{C})$ *format*=*textbf* *classofobjects* *of*category.

- For each pair of objects $X, Y \in \text{Ob}(\mathcal{C})$, a class *classofmorphismsbetween* *two* *objects* *of*categoryname = description=Class of morphisms between objects X and Y of the category $\mathcal{C}$ (??), sort=Hom,

$$\text{Hom}_{\mathcal{C}}(X, Y)$$

format=*textbf* *classofmorphismsbetween* *two* *objects* *of*category *of* *morphism* *between* *objects* *of*category *format*=*textbf* *morphism* *between* *objects* *of*category (also called *arrow* *between* *objects* *of*categoryname = *Arr* *format*=*textbf* *arrow* *between* *objects* *of*category or *hom* *between* *objects* *of*categoryname = *Hom* *between* *objects* *of*category) *format*=*textbf* *hom* *between* *objects* *of*category). If the category $\mathcal{C}$ is clear, then this is also denoted by. It may also

- For each triple of objects X, Y, Z , a composition law

$$\circ : \text{Hom}_{\mathcal{C}}(Y, Z) \times \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{C}}(X, Z),$$

denoted $(g, f) \mapsto g \circ f$.

- For each object X , an *identity morphism*

$$\mathrm{id}_X \in \mathrm{Hom}_{\mathcal{C}}(X, X).$$

These data satisfy the following axioms:

- (Associativity) For all morphisms $f \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$, $g \in \mathrm{Hom}_{\mathcal{C}}(Y, Z)$, and $h \in \mathrm{Hom}_{\mathcal{C}}(Z, W)$,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

- (Identity) For all $f \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$,

$$\mathrm{id}_Y \circ f = f = f \circ \mathrm{id}_X.$$

One often writes $X \in \mathcal{C}$ synonymously with $X \in \mathrm{Ob}(\mathcal{C})$, i.e. to denote that X is an object of $\mathcal{C}$.

We may call a category as above an *ordinary category* to distinguish this notion from the notions of *categories enriched in monoidal categories* or higher/ n -categories. (♠ TODO: define n -categories)

A category as defined above may be called called a *large category* or a *class category* to emphasize that the hom-classes may be proper classes rather than sets (note, however, that the possibility that hom-classes are sets is not excluded for large categories). Accordingly, a *category* may often refer to a locally small category, which is a category whose hom-classes are all sets.

Definition B.0.2 (Locally small category). A (large) category $\mathcal{C}$ is called a *locally small category* if for every pair of objects $X, Y \in \mathrm{Ob}(\mathcal{C})$, the collection $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ of morphisms between them is a (small) *set* (as opposed to a proper class). In other words, each hom-class is a set and may even be called a *hom-set*.

In some contexts, a locally small category may simply be called a *category*, especially when genuinely large categories are not considered.

A category $\mathcal{C}$ is called a *small category* if it is a locally small category and the class $\mathrm{Ob}(\mathcal{C})$ of objects is a set.

Remark B.0.3. Many “concrete” categories considered in “classical mathematics” or outside of more “abstract” category theory tend to be locally small. For example, the categories of sets, groups, R -modules, vector spaces, topological spaces, schemes, manifolds, sheaves on “small enough” sites are all locally small.

APPENDIX C. GROUP THEORY

Definition C.0.1 (Groups). A *group* is a pair $(G, \cdot)$ where G is a set and $\cdot : G \times G \rightarrow G$ is a binary operation, subject to the following conditions:

1. (Associativity) For all $g, h, k \in G$ one has

$$(g \cdot h) \cdot k = g \cdot (h \cdot k).$$

2. (Identity element) There exists an element $e \in G$ such that for all $g \in G$,

$$e \cdot g = g \cdot e = g.$$

3. (Inverse element) For all $g \in G$ there exists an element $g^{-1} \in G$ such that

$$g \cdot g^{-1} = g^{-1} \cdot g = e.$$

The element e is called the *identity element of G* , and g^{-1} is called the *inverse of g* .

Equivalently, a group is a monoid with inverse elements.

A group $(G, \cdot)$ is often simply written as G , when the notation for the binary operation $\cdot$ is clear.

An *abelian group* or synonymously, a *commutative group*, is a group $(G, \cdot)$ whose binary operation $\cdot$ is *abelian* or *commutative*, i.e. satisfies

$$g \cdot h = h \cdot g$$

for all $g, h \in G$.

Definition C.0.2 (Group homomorphism). Let $(G, \cdot)$ and $(H, *)$ be groups. A map $f : G \rightarrow H$ is called a *group homomorphism* if for all $g_1, g_2 \in G$ one has

$$f(g_1 \cdot g_2) = f(g_1) * f(g_2).$$

The collection of all groups with the group homomorphisms forms a locally small category, called the *category of groups*.

If f is bijective, then f is called a *group isomorphism*.

APPENDIX D. RING THEORY

Definition D.0.1. A *commutative (unital) ring* is a ring $(R, +, \cdot)$ such that $\cdot$ is a commutative operation, i.e. $a \cdot b = b \cdot a$.

For many writers (e.g. “commutative” algebraists or number theorists), a *ring* refers to a commutative ring as above.

Definition D.0.2. Let R be a not-necessarily commutative ring.

1. A *left R -module* is an abelian group $(M, +)$ together with an operation $R \times M \rightarrow M$, denoted $(r, m) \mapsto rm$, such that for all $r, s \in R$ and $m, n \in M$:
 - $r(m + n) = rm + rn$,
 - $(r + s)m = rm + sm$,
 - $(rs)m = r(sm)$,
 - $1_R m = m$ where 1_R is the multiplicative identity of R .
2. A *right R -module* is defined similarly as an abelian group $(M, +)$ with an operation $M \times R \rightarrow M$, denoted $(m, r) \mapsto mr$, such that for all $r, s \in R$ and $m, n \in M$:
 - $(m + n)r = mr + nr$,

- $m(r + s) = mr + ms$,
 - $m(rs) = (mr)s$,
 - $m1_R = m$.
3. Let R and S be (not necessarily commutative) rings.
 An R - S -bimodule (or an R - S -module or an (R, S) -module, etc.) is an abelian group $(M, +)$ equipped with
 (a) a left action of R :

$$R \times M \rightarrow M, \quad (r, m) \mapsto r \cdot m,$$

- making M a left R -module,
 (b) a right action of S :

$$M \times S \rightarrow M, \quad (m, s) \mapsto m \cdot s,$$

- making M a right S -module,
 such that the left and right actions commute; that is, for all $r \in R$, $s \in S$, and $m \in M$,

$$r \cdot (m \cdot s) = (r \cdot m) \cdot s.$$

4. A $two\text{-sided } R\text{-module}$ (or R -bimodule) is an R - R -bimodule.

If R is a commutative ring, then a left/right R -module can automatically be regarded as a two-sided R -module. As such, we simply talk about R -modules in this case.

Any abelian group is equivalent to a two-sided $\mathbb{Z}$ -module. Moreover, any left R -module is equivalent to an $R - \mathbb{Z}$ -bimodule and any right R -module is equivalent to an $\mathbb{Z} - R$ -bimodule. Given a left/right/two-sided R -module, its *natural bimodule structure* will refer to its structure as a $R - \mathbb{Z} / \mathbb{Z} - R / R - R$ bimodule. In this way, many definitions associated with the notions of left/right/two-sided R -modules can be defined as special cases for definitions for R - S -bimodules.

Definition D.0.3 (Hom of left/right/bi-modules). Let R, S, T be (not necessarily commutative) rings.

1. Let M and N be left R -modules. The *homomorphism group of left R -modules from M to N* is the abelian group

$$\text{Hom}(M, N) = \text{Hom}_R(M, N) = \text{Hom}_{R-\mathbb{Z}}(M, N) := \{f : M \rightarrow N \mid f \text{ is a left } R\text{-module homomorphism}\}.$$

2. Let M and N be right R -modules. The *homomorphism group of right R -modules from M to N* is the abelian group

$$\text{Hom}(M, N) = \text{Hom}_R(M, N) = \text{Hom}_{\mathbb{Z}-R}(M, N) := \{f : M \rightarrow N \mid f \text{ is a right } R\text{-module homomorphism}\}.$$

3. Let S be a (not necessarily commutative ring) and let M and N be $R - S$ -bimodules. The *homomorphism group of R - S -bimodules from M to N* is the abelian group

$$\text{Hom}(M, N) = \text{Hom}_{R-S}(M, N) := \{f : M \rightarrow N \mid f \text{ is a } R - S\text{-bimodule homomorphism}\}$$

In each case, $\text{Hom}(M, N)$ has a natural structure of an **abelian group** given by **pointwise addition**: for $f, g \in \text{Hom}(M, N)$,

$$(f + g)(m) := f(m) + g(m),$$

and the zero morphism $\mathbf{0}$ given by $0(m) := 0_N$ acts as the identity element. The additive inverse $-f$ is defined by $(-f)(m) := -f(m)$. Moreover, depending on bi-module structures that M and N may be carrying, $\text{Hom}(M, N)$ may itself carry additional module structures:

- In case that M is a $R - S$ -bimodule and N is a $R - T$ -bimodule, $\text{Hom}_R(M, N)$, the group of left R -module homomorphisms, is an $S - T$ -bimodule as follows:

$$(s \cdot f \cdot t)(m) = f(m \cdot s) \cdot t \quad f \in \text{Hom}_R(M, N), s \in S, t \in T.$$

- Dually, in case that M is a $S - R$ -bimodule and N is a $T - R$ -bimodule, $\text{Hom}_R(M, N)$, the group of right R -module homomorphisms, is an $S - T$ -bimodule as follows:

$$(s \cdot f \cdot t)(m) = f(s \cdot m) \cdot t \quad f \in \text{Hom}_R(M, N), s \in S, t \in T.$$

Some cases of interest may be when R, S , or T is in fact $\mathbb{Z}$ — these allow us to see module structures on $\text{Hom}(M, N)$ even when M and N are one-sided modules.

(♠ **TODO: state this as a theorem**) We furthermore note that $\text{Hom}_R(-, -)$ yields biadditive functors

$$\text{Hom}_R(-, -) : {}_R\mathbf{Mod}_S^{\text{op}} \times {}_R\mathbf{Mod}_T \rightarrow {}_S\mathbf{Mod}_T$$

$$\text{Hom}_R(-, -) : {}_S\mathbf{Mod}_R^{\text{op}} \times {}_T\mathbf{Mod}_R \rightarrow {}_S\mathbf{Mod}_T.$$

APPENDIX E. HOMOLOGICAL ALGEBRA

Definition E.0.1 (Chain complex in a preadditive category). Let $\mathcal{A}$ be a preadditive category and let I be a totally ordered set (typically $\mathbb{Z}$, but $I \subseteq \mathbb{Z}$ is also allowed).

1. A **chain complex** $(K_\bullet, d_\bullet)$ in $\mathcal{A}$ indexed by I is the homological convention for sequences with decreasing degrees. It consists of:

- Objects $\{K_i\}_{i \in I}$ in $\mathcal{A}$, called the **terms in degree i** ,
- Morphisms $d_i : K_i \rightarrow K_{i-1}$ in $\mathcal{A}$, called the **boundary maps** or **differentials in degree i** ,

such that for every $i \in I$, $d_{i-1} \circ d_i = 0$. That is,

$$K_\bullet : \cdots \xrightarrow{d_{i+1}} K_i \xrightarrow{d_i} K_{i-1} \xrightarrow{d_{i-1}} K_{i-2} \rightarrow \cdots$$

with $d_{i-1}d_i = 0$ for all i . We typically use the notation $K_\bullet = (K_i, d_i)_{i \in I}$.

2. Dually, a **cochain complex** $(K^\bullet, d^\bullet)$ in $\mathcal{A}$ follows the cohomological convention with increasing degrees. It consists of objects $\{K^i\}_{i \in I}$ and **coboundary maps** $d^i : K^i \rightarrow K^{i+1}$ such that $d^{i+1} \circ d^i = 0$:

$$K^\bullet : \cdots \xrightarrow{d^{i-1}} K^i \xrightarrow{d^i} K^{i+1} \xrightarrow{d^{i+1}} K^{i+2} \rightarrow \cdots$$

We typically use the notation $K^\bullet = (K^i, d^i)_{i \in I}$.

3. Let $K_\bullet = (K_i, d_i^K)$ and $L_\bullet = (L_i, d_i^L)$ be chain complexes in $\mathcal{A}$ indexed by the same set I . A **morphism of chain complexes** (or **chain map**)

$$f_\bullet : K_\bullet \rightarrow L_\bullet$$

consists of morphisms $f_i : K_i \rightarrow L_i$ for all $i \in I$, such that for every $i \in I$, the following diagram commutes:

$$\begin{array}{ccc} K_i & \xrightarrow{d_i^K} & K_{i-1} \\ \downarrow f_i & & \downarrow f_{i-1} \\ L_i & \xrightarrow{d_i^L} & L_{i-1} \end{array}$$

i.e., $d_i^L \circ f_i = f_{i-1} \circ d_i^K$.

A **morphism of cochain complexes** $f^\bullet : K^\bullet \rightarrow L^\bullet$ is defined similarly, satisfying the commutativity condition $d_L^i \circ f^i = f^{i+1} \circ d_K^i$.

The collection of these objects and morphisms forms a category. Notation for these categories is as follows:

- $\mathbf{Ch}(\mathcal{A})$ or $\mathbf{Ch}(\mathcal{A})$ is often used as a general term.
- To be explicit about the indexing convention, one uses $\mathbf{Ch}_\bullet(\mathcal{A})$ for chain complexes and $\mathbf{Ch}^\bullet(\mathcal{A})$ (or sometimes $\mathbf{CoCh}(\mathcal{A})$) for cochain complexes.
- The set of chain maps between two complexes is denoted by $\mathbf{Hom}_{\mathbf{Ch}(\mathcal{A})}(K_\bullet, L_\bullet)$; it is an abelian group under pointwise addition $(f + g)_i = f_i + g_i$.

Definition E.0.2 (Chain complexes and their (co)homology objects). Let $\mathcal{A}$ be an abelian category.

- For a cochain complex $K^\bullet$, its **cohomology object in degree i** is defined as the quotient of the object of i -cocycles by the object of i -coboundaries:

$$H^i(K^\bullet) := Z^i(K)/B^i(K) = \ker(d^i)/\mathrm{im}(d^{i-1}).$$

In fact, each H^i is a functor $H^i : \mathbf{Ch}^\bullet(\mathcal{A}) \rightarrow \mathcal{A}$; given a morphism $f^\bullet : K^\bullet \rightarrow L^\bullet$ of cochain complexes, it is thus customary to write $H^i(f^\bullet) : H^i(K^\bullet) \rightarrow H^i(L^\bullet)$ for the induced morphism on the i th cohomology objects.

- For a chain complex $K_\bullet$, its **homology object in degree i** is defined as the quotient of the object of i -cycles by the object of i -boundaries:

$$H_i(K_\bullet) := Z_i(K)/B_i(K) = \ker(d_i)/\mathrm{im}(d_{i+1}).$$

In fact, each H_i is a functor $H_i : \mathbf{Ch}_\bullet(\mathcal{A}) \rightarrow \mathcal{A}$; given a morphism $f_\bullet : K_\bullet \rightarrow L_\bullet$ of chain complexes, it is thus customary to write $H_i(f_\bullet) : H_i(K_\bullet) \rightarrow H_i(L_\bullet)$ for the induced morphism on the i th homology objects. .

See ??

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